

Assignment 5: Pacman Protocol

Design and Specification

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ENGF0034 – Design and Professional Skills

Lecture Overview

Objectives

This session introduces the "Pacman Protocol Specification" assignment. We will review the context of the multi-player game, analyse the flaws in the current networking implementation, and define the requirements for designing a robust, secure, and language-independent protocol specification.

Lecture Overview

- 1 Introduction and Context
- 2 The Current Implementation
- 3 Specification Requirements
- 4 Submission and Marking

Introduction and Context

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The Assignment

The Goal

Your task is **not** to write code. Instead, you must write a technical **Protocol Specification** for a multi-player networked Pacman game.

Why Specification?

Writing code is only half the job. In professional engineering, the ability to communicate *how* a system works—unambiguously and concisely—is critical for interoperability and maintenance.

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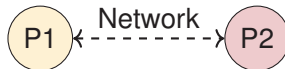
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The Game Scenario

We have extended the classic Pacman game to support two players.

- **Original Intent:** A sociable game played by two people in the same room.
- **Mechanism:** If your Pacman enters the tunnel, it disappears from your screen and appears on your opponent's screen.
- **Remote Mode:** Added during the pandemic to allow play across the internet, displaying a "mini-view" of the opponent's screen.



Game Terminology

To specify the protocol, we must agree on terminology regarding where objects are.

LOCAL: A game object currently on your own screen.

AWAY: Your Pacman when it has travelled to the remote screen.

REMOTE: A game object (like a ghost) on the remote screen that your **AWAY** Pacman interacts with.

FOREIGN: The opponent's Pacman when it is visiting your screen.

Implication

Your local model must track both the **Local Maze** and the **Remote Maze** to handle interactions efficiently.

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The Current Implementation

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The Existing Protocol

The current alpha-release contains a working network protocol, but it is deeply flawed.

Origin Story

Implemented by an anonymous professor late at night after a pub visit. It is a "quick and dirty" solution.

Major Flaws

- 1 **Language Dependent:** Uses Python's 'pickle' module.
- 2 **Insecure:** 'pickle' is vulnerable to malicious input (arbitrary code execution).
- 3 **Ugly Encoding:** A mix of binary length headers, text types, and pickled payloads.
- 4 **Inefficient:** "Chatty" behaviour (e.g., sending four separate ghost updates per frame).

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Why 'pickle' must go

- **Interoperability:** A protocol using 'pickle' can only be spoken by Python programs. A standard protocol should be implementable in C++, Java, Rust, etc.
- **Security:** Deserialising untrusted 'pickle' data is dangerous.

Your Challenge

Design a replacement protocol that is **clean**, **secure**, and **language-agnostic**.

Specification Requirements

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Core Constraints

Your specification must adhere to the following strict constraints:

1. Encoding Format

- You must choose **Text-based** (e.g., ASCII commands) OR **Binary-based** encoding.
- Do **not** mix them without a compelling documented reason.
- **Forbidden:** You may NOT use JSON, XML, HTML, or 'pickle'. You must design the byte-level or character-level format yourself.

2. Transport Layer

- You may use **TCP**, **UDP**, or a combination of both.
- If using UDP, you must specify how to handle packet loss (e.g., retransmission or ignoring outdated state).

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Scope of Messages

You must specify how to encode information for all game events. The current protocol uses 12 message types:

- | | |
|-------------------------------------|------------------------------------|
| 1 Maze update (sent at start) | 7 Ghost update |
| 2 Pacman arrived (tunnel transit) | 8 Ghost was eaten |
| 3 Pacman left | 9 Foreign pacman ate ghost |
| 4 Pacman died | 10 Eat (food/powerpill) |
| 5 Pacman go home | 11 Score update |
| 6 Pacman update (position/velocity) | 12 Status update (Game Over, etc.) |

Note: Your new protocol does not need to map these 1:1, but it must cover all functionality.

Sanity Checking and Robustness

A key part of a specification is defining validity.

Defensive Design

Your specification must indicate how received values should be checked.

- **Example:** If a message claims "Ghost #5" moved, but there are only 4 ghosts, what should happen?
- **Example:** If a coordinate is outside the maze bounds?
- The specification must be unambiguous enough for another programmer to implement these checks.

Submission and Marking

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Deliverables

Submission

- Submit via Moodle.
- This document is a technical specification, not a story.
- No code submission is required for this assignment.

Peer Marking

You will mark the specifications of your peers within your group. **Mental Model:** When marking, ask yourself: *"Could I implement the game networking using ONLY this document, without guessing?"*

Marking Criteria

You should optimise your specification for the following qualities:

Conciseness: Be specific. Do not "waffle". This is technical documentation, not literature.

Correctness: Will the protocol actually work? Are there logical flaws?

Unambiguity: Is there only one way to interpret the byte layout?

Completeness: Are edge cases (like Game Over or connection loss) handled?

Examples

You may use examples to clarify complex parts (non-normative), but they should not replace the formal definition.

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Getting Started

- 1 **Run the Game:** Download the alpha code. Run the server and client (or relay server). Play it (use `-r` for remote mode).
- 2 **Inspect the Code:** Look at `pa_network.py` and `pa_model.py`.
- 3 **Debug:** Add `print()` statements to `check_for_messages()` to see what the current bad protocol is actually sending.
- 4 **Design:** Draft your message structures on paper before writing the spec.

Warning

The assignment document describes the game, but if there is a conflict between the document and the Python code behaviour, **the code is authoritative**.

Summary I

Key Takeaways

- **Task:** Design a replacement network protocol for Pacman.
- **Constraint:** No 'pickle', no JSON/XML. Consistent text or binary encoding.
- **Scope:** Handle all game states (Movement, Eating, Tunnels, Game Over).
- **Goal:** Produce a complete, unambiguous specification document.
- **Deadline:** Check Moodle.